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Lab Report

Switch Capacitor DC-DC Converter design

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In modern mixed signal SoC, we often need to generate different voltage from a single battery supply. Linear regulators have the advantage of low area and large current. However in modern SoC, some digital circuits operate in a voltage as low as 0.9V, which comes from a 1.8V power supply. This 0.9V voltage difference makes linear regulator inefficient. Hence switched capacitor dc-dc converter is better for large voltage drop conversion.

There are three important criteria. First, the size of switched capacitor need to be samll enough to fully integrate on the chip. Second, the size should be close to linear regulators. Third, the noise with switching operation must be predictable.

The technology we use in this essay is 180nm. This technology provides large MIM capacitors. Which is one way to achieve our voltage under large load current. Another way is to use 22nm technology, smaller technology allows us to further increase the clock (due to smaller parasitic capacitance). Which will not be discussed in this essay.

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1.1 Spec

Table 1.1: Specifications(original)

Input voltage	3.3V
Output voltage	1.8V
Process	180nm
Average load current	1mA
Peak load current	5mA
CLK peak	100M Hz
Sleep mode clock	32.768kHz
Sleep mode current	1uA

1.2 Equations

Parameter	Equation
R_{SSL}	$\sum_{i \in \text{Caps}} \sum_{j=1}^n \left(\frac{\alpha_{c,i}^{j2}}{C_i f_{sw}} \right)$
R_{FSL}	$\sum_{i \in \text{switch}} \sum_{j=1}^n \left(\frac{\alpha_{r,i}^{j2} \cdot R_i}{D_j} \right)$
R_{out}	$\sqrt{R_{SSL}^2 + R_{FSL}^2}$
Output ripple	$\frac{\Delta V}{2}$
K_{SSL}	$\sum_{i=1}^{n_{ph}} \sum_{j=1}^n \alpha_{c,j}^i$
K_{FSL}	$\sum_{i=1}^{n_{ph}} \sum_{j=1}^n \alpha_{r,j}^i$

1.3 Basic introduction

1.3.1 Switch capacitor topology introduction

There are 4 mainly used topology, series-parallel, ladder, Fibonacci, Dickson. For further information and realizable conversion ratio please refer to [1].

1.3.2 Basic current flow: Ladder 5:3

We take 5:3 Ladder converter for example and visualize the current flow. 1.1 shows the charge transfer in a 5:3 ladder converter. From power perspective $P_{in} = P_{out}$, during phase 1 P_{in} draws $0.6Q_{in}$ from V_{in} , and is disconnected during phase 2. Output capacitor receive a total amount of Q_{in} charge. Total power draw from input = $0.6Q_{in} * 3.3V = Q_{in} * 0.6 * 3.3V = Q_{out} * V_{out}$

1.3.3 Rout

R_{out} is important because it directly affects our output voltage drop as well as our efficiency by dissipating power. Note that after determining topology and capacitance. R_{SSL} comes from the charge loss during the charging/dischARGE process. This will cause voltage ripple in flying capacitance, then affect our output. Thus to reduce this we can either make capacitance larger or make frequency larger. (inversely proportional to C_{fly} , f_{sw})

R_{FSL} is inversely proportional to W . It arise when flying capacitance voltage is nearly constant. Which happens when we have high RC constant or our frequency is fast. Then we need to consider more the power dissipation in resistor when conducting.

1.3.4 Clock non-overlap circuit

If we don't have this, there will be a time that both our PMOS and NMOS is conducting, which will affect efficiency.

1.3.5 Gate drive

The design of gate driver can be found in [1]. Critical is that the gate driver should be able to drive our switches without consuming too much power.

1.3.6 The comparison of different topology

[2] [3] discussed different topologies. There are 4 common topologies: Ladder, Dickson, Fibonacci and Series-Parallel. We can conclude that some of them are better when it comes to large step down ratio. However in low step down ratio such as 3:1, the difference is small. I will also introduce another reconfigurable topology in [4]. Which is a basic cell that is often used to create multiple conversion ratio.

1.3.7 Output voltage ripple

Large ripple might consume extra power according to [5]. Output voltage ripple can be reduced by using multiple interleaved phases, without the need to increase the size of output capacitance C_{out} . Since switch capacitor converter is mostly used on chip. There's a trade off between not using too large output capacitor and keeping ripple small.

1.3.8 Sleep mode control

Sleep mode is common in ULV(ultra low voltage) applications to reduce power consumption and increase battery life. Sleep mode needs to remain same voltage but

operating in a much lower load current.

1.4 Design Step

1. Set a target efficiency (e.g., 50%), and be aware of how much capacitance (area) we are allowed to use.
2. Determine the load current we want to optimize
3. Calculate load power and the amount of power we are able to dissipate.
4. Build the converter and test if capacitance, switch size, and frequency are optimized. Verify if we are able to achieve the desired output under load current.
5. If not, consider changing the topology.

Table 1.2: Key Equations

Equation Name	Equation
Average Load Power	$(V_{\min} + \frac{1}{2}\Delta V) \cdot (I_L + \frac{1}{2}\Delta I)$
Intrinsic loss	$I_{load}^2 R_{out}$
SSL Loss	$I_{load}^2 R_{SSL}$
Ripple Loss	$I_L \cdot \frac{1}{2}\Delta V$
Gate control loss	$C_{gate} V_{drv}^2 f_{sw}$
Switch Loss	$I_L^2 R_{on} \cdot (\text{Num. of switch})$
Bottom capacitor loss	$\frac{1}{2} V_{r, \text{cap}}^2 C_{\text{bott}} f_{sw}$
Control loss	Loss of digital control blocks
Efficiency	$\frac{P_{\text{out}}}{P_{\text{in}}}$

1.5 Design 1: Ladder 5:3 converter

1.5.1 Comparator design

I use a strong arm comparator in [6]. Critical for comparator is that it's low power, it's able to drive next stage, and it has small offset. To compensate offset we add capacitance.

[7] provide detailed analysis for comparator, and state that V_{cm} is critical for speed because it determines the discharge in input phase. Fortunately in our design the V_{cm} is 1.8. Provides great speed.

1.5.2 Advantages

This topology is useful since it can provide multiple stable DC levels. Which can be used as high and low bound of our gate driver. This is not implemented in my design because I have sleep mode. In this mode I want to completely cut off the leakage current of the MOSFET, thus I prefer to let gate discharge to 0 rather than discharge to source voltage.

1.5.3 Challenges

This ratio is too tight for 1.8V. We can't even deliver 50uA load current and maintain 1.8V. Another drawback is that we use lots of capacitance in this design, which consumes large area. In order to solve them, I tried another topology.

1.6 Design 2: 5:3 converter

1.6.1 Converter design

See 1.3 for building blocks.

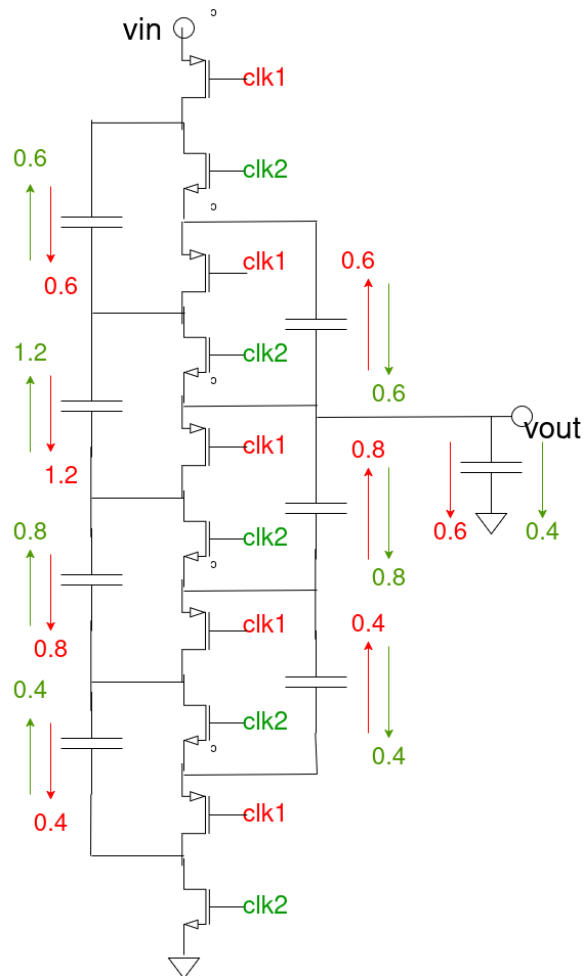


Figure 1.1: Current flow of ladder 5:3 converter

1.6.2 Result

We can reach efficiency of 50% under $100\mu\text{A}$ load current.

1.6.3 Challenges

To some extent we cannot optimize it more. Either due to capacitance area constraint, or due to the need to deliver large load current. The end of our optimization will be - same value for both R_{SSL} and R_{FSL} , minimum sum of gate drive loss and switch conduction loss.

Table 1.3: Performance design 2

Parameter	Value
Input voltage	3.3V
Output voltage	1.8V
Process	180nm
Output capacitance	200pF
Flying capacitance	50pF
Total capacitance	350pF
Efficiency	50%

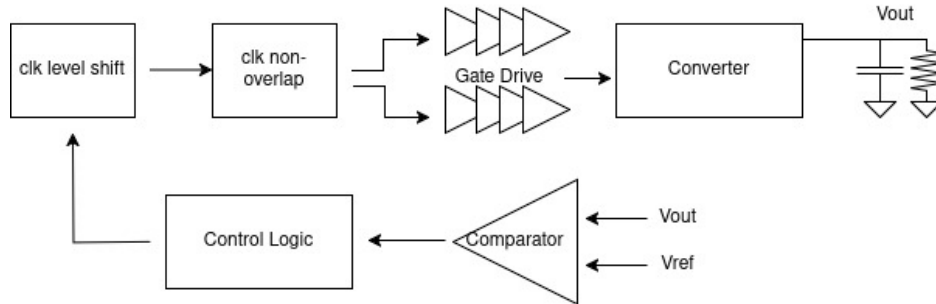


Figure 1.3: Top level

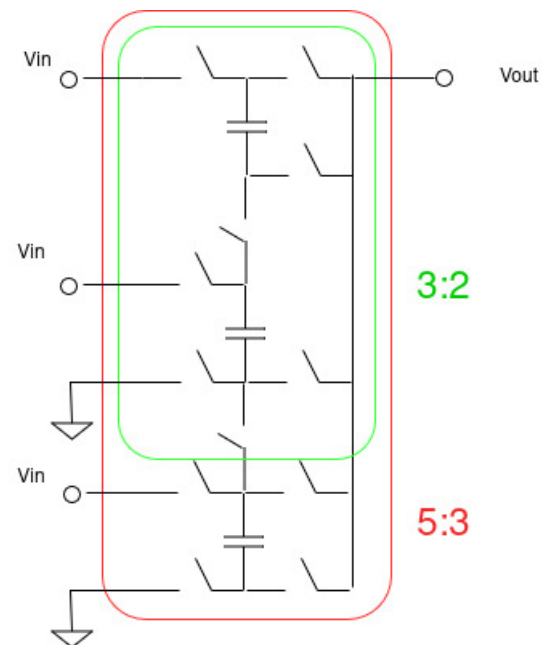


Figure 1.2: Double ratio Converter

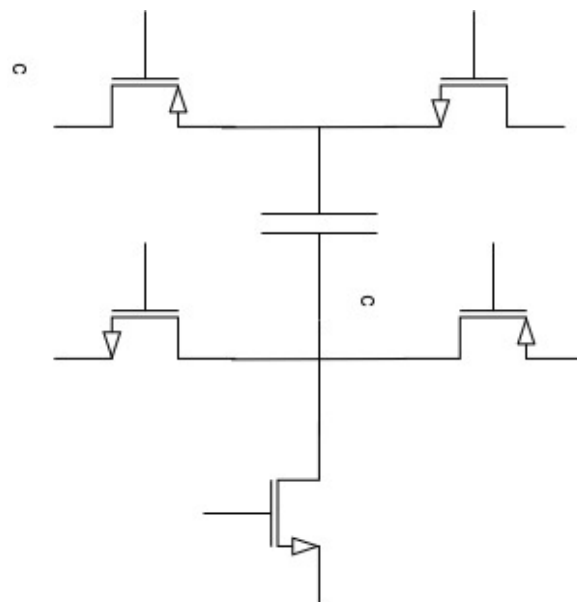


Figure 1.4: Basic cell

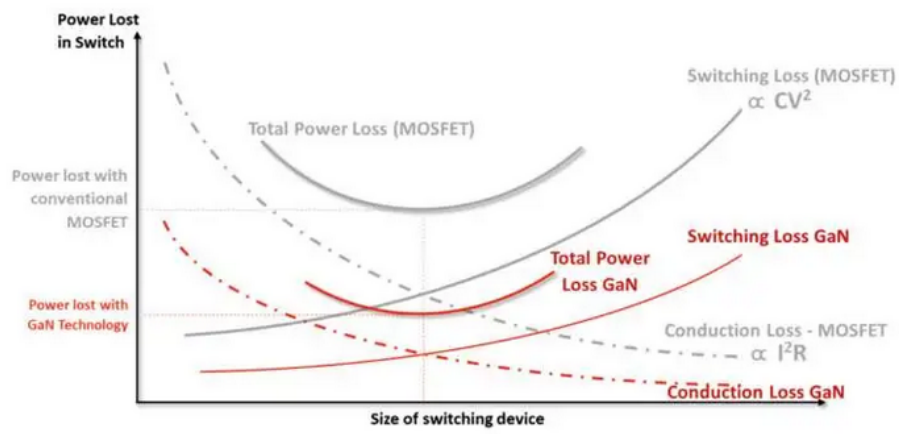


Figure 1.5: Loss curve

1.7 Design 3: 3:2 converter

In order to provide 1mA load current, we can only use 3:2 topology. Besides, 3:2 topology also has better efficiency. This is because this topology has fewer switch. Although having higher intrinsic loss, compared to switch conduction loss it's still small.

1.7.1 Clock shifting optimization

Clock shifting is a technique in [4]. It use 2 different clock level to decrease the switch gate loss. I'm able to reach 76% of efficiency in normal operation mode at 1.8mW.

1.7.2 Challenges

After achieving 70% efficiency, I try to optimize the design by adjusting frequency to make SSL equal to FSL. However when we decrease frequency, our capacitance needs to be bigger to provide 1mA load current. The trade off here is that I get 2% improvement by making capacitance double. Which at this point I think it's not worth it. For comparison please see 1.5.

1.7.3 Sleep mode challenge

Our original sleep mode clock 32.768k Hz and sleep mode current $1\mu\text{A}$ is hard to achieve. Since the slow clock will cause large ripple and will cause large power loss in our switch. Low load current will let our load capacitance discharge slower. Thus there are two way to increase efficiency. Either make both frequency and load current larger. Or, try to find a method to decrease ripple.

In this work I try to decrease the conductivity of my switch. Adjust the body bias can decrease current, but it's hard to control since the resistance change drastically also non-linear and it will burn lots of power. Changing the length is a better method.

After increasing the frequency I'm able to reach 58% efficiency. See table 1.6.

Table 1.4: Performance design 3

Input voltage	3.3v
Output voltage	1.8V
Process	180nm
Output capacitance	200p
Flying capacitance	50p
Total capacitance	300p
Efficiency 1.8mW	70%
Power density	dd

1.8 Loss optimization

1.8.1 Spec

Before we start optimization, we need to know what our load current is. We always optimize under certain load current value. It determines the minimum switch size and capacitance size we require.

1.8.2 Frequency optimization

The frequency optimization formula, according to [1], is

$$\left(\frac{K_{SSL}}{K_{FSL}} \right)^2 \times \frac{G_{tot}}{2C_{tot}}$$

This formula comes from the point where FSL and SSL is equal to get smallest Rout. If we know the limit of capacitance that we can use. We can optimize our frequency.

Here I take design 1 for example, suppose our flying capacitance limit is 100p, the G of our switch is 1/300, K_{SSL} is 3/5, K_{FSL} is 11/5. We can easily get optimal frequency 9.1×10^6 which is close to 10^7 .

1.8.3 Switch loss

There are conduction loss and switching loss, in order to reduce conduction loss we have to make the size of our switch big. But when width size of switch is above 5μ , making it bigger doesn't reduce resistance a lot. (According to simulation result the decrease of switch resistance saturate) On the other hand it will consume a lot of power in switching loss, because we have large gate capacitance which charge and discharge slow and will burn power. Thus we cannot make it bigger than 5μ .

The problem is, what is the optimum size of our switch?

1.8.4 Gate Loss curve

See 1.5, we try to find intersection point between two curves. There's the limit, if we already size the switch as large as we can, at the same time reduce frequency to ensure that our flying capacitance is fully charge/discharged, still we cannot reach 1.8V. At the same time our flying capacitance is also at limit size. We have to change the transfer ratio or try other topology. This is also the reason I abandon design 1: Ladder 5:3 converter.

Shortly speaking, we calculate the loss of every single switch, make the switch that burns more power bigger. Constantly changing the switch size and find a minimum sum of gate loss and conduction loss.

1.8.5 Ripple loss

When we have small load capacitance, we will have large ripple. According to my calculation, ripple loss is not very critical in normal operation mode. But, if our load current is very small such as in sleep mode, every ripple will cause current flow in our switch and is significant. We can use techniques such as interleaving to reduce our ripple. Connect the body of our switch to either VDD or VSS to change the conductivity is not a good idea, since we will increase the resistance leads to larger power consumption.

1.8.6 Bottom plate loss

If we increase the flying capacitance, the bottom plate parasitic will increase, the frequency as well as the rail to rail voltage of cap will decrease. Lead to a overall decrease in parasitic loss. This is often ignored.

1.8.7 Leakage loss

In order to provide large current at normal mode, we need to make switch large. This will cause a huge leakage current when we operate in sleep mode.

1.9 Control

- **Pulse Skip Modulation (PSM):** [8] explained the PSM circuit. The charge transfer stops when V_{out} exceeds the desired reference voltage V_{ref} .
- **Pulse Frequency Modulation (PFM):** [9] introduced PFM in detail. If the output voltage falls below V_{ref} , the switching frequency is increased, and the switch size is enlarged. The paper discusses using three frequency levels: f_{min} , $2f_{\text{min}}$, and $4f_{\text{min}}$.
- **Capacitance Modulation:** While PSM and PFM are stable and frequently used, they introduce switching noise at different frequencies. [10] proposed capacitance modulation to maintain a fixed frequency. However, this method requires a large capacitance and results in high power consumption in the control circuit. To mitigate this, the paper suggests switching to PSM when the load current is stable and low.

1.9.1 Interleaving

Interleaving is critical for low frequency sleep mode. To reduce the ripple and reduce power consumption. See [10]

1.9.2 Conclusion pre-layout simulation

When designing high voltage converter. We need to use 3.3V or even 5V MOSFET. These MOSFET have big size as well as large gate capacitance. This limits the speed of our digital circuit. Also require larger gate drive. We also need to deal with mismatches in comparator by adding capacitance.

1.10 Comparison

Work	This work	[11]	[4]	[10]
Topology	3/2 step down	2/1 step down	3/2 2/1 3/1 step down	3/2 step down
C_{fly} total	100pF	1320pF	unknown	534pF
C_{load}	200pF	2nF	0	700pF
Technology	180nm	28nm	32nm	45nm
Efficiency	76%	82%	79.7%	69%
Load	1.8mW	3.5mW	unknown	8mA
Power density	17	18	$0.86W/mm^2$	$50mW/mm^2$
Area	17	18	19	20

Table 1.5: Comparison normal mode

1.11 Layout

Work	This work	[11] Design 1	[11] Design 2
Topology	3/2 step down	2/1 step down	2/1 step down
C_{fly} total	100pF	200pF	1320pF
Technology	180nm	130nm	28nm
Efficiency	57.8%	63%	82%
CLK Frequency	200kHz	65kHz	13MHz
Load	1.8uW	100nW	40uW

Table 1.6: Comparison sleep mode

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